The Inventory System

Parts, what each entity needs, who owns what is on hand, and where it is.

GREAT CIRCLE SOLAR OPERATIONS
INVENTORY SYSTEM · PARTS AND STOCK

The inventory system keeps the list of spare parts, how many of each every entity needs, who owns the ones on hand, and where they physically are. It also holds the purchase order process, which has its own guide.

> **WHO CAN CHANGE THINGS**
> Everyone who is signed in can **see** the Master List, Required Inventory, Ownership, Physical Location and History. Only people with the **Inventory** role can **change** any of them. If you do not have it you will not see the edit buttons. Ask an admin if you need it.

The tabs

TAB	WHAT IT IS	WHO SEES IT
Master List	Every part the company knows about, once, with its GCS P/N.	Everyone signed in (edit: **Inventory**)
Required Inventory	For each entity: which parts it needs and how many it should keep (target stock).	Everyone signed in (edit: **Inventory**)
Inventory On Hand	What we actually have. Three sub-tabs: **Ownership**, **Physical Location** and **History**.	Everyone signed in (edit: **Inventory**)
Purchase Orders	See the Purchase Order guide.	Everyone, vendors included
PO History	Tamper-proof record of every PO change.	Admin and PO roles
Audit	Compares inventory against accounting's numbers. **Still a preview**: it opens on made-up data and saves nothing.	Admin only for now
Budget	Budget against spending, by entity and category.	Admin only for now
Admin	Users, vendors, projects, entities, sites, categories and settings.	**Admin**
Suggestions	Tell the admin about something broken, disliked, or an idea.	Everyone except vendor logons

The search box at the top right of the page finds POs and parts. Typing a GCS P/N takes you to that part.

The ideas that explain everything else

- **Owned** and **where it is** are two separate numbers. **Ownership** says which entity's count a unit belongs to. **Physical Location** says where the unit is sitting. For every part the two totals always match; the system refuses any change that would break that.
- Places a unit can be: **Oshawa**, the **Barn**, the **Office**, or an entity's **on-site shelf** (shown as *<Entity> (on site)*).
- An entity's on-site shelf can only hold parts **that entity owns**. To move a part from one entity's shelf to another's, transfer ownership first (Stock Transfer), which moves the units with it.
- Some entities keep no parts on an on-site shelf (Firelight Operations and Firelight Solar). They still own parts, which are kept in Oshawa, the Barn or the Office, and they have no on-site column on Physical Location.
- Stock is never edited directly. Each change goes through a check that applies the rules and writes a line in **History**, so there is always a record of who did what and why. A reason is required for every change.
- Yellow numbers are spares on **issued purchase orders that have not been received yet**. They are counted so you can see what is coming, but they are not on the shelf until the requisitioner marks the PO's parts Received.

> **WORDS**
> Three layers, from the top down. A **Project** is the top layer (Regular O&M, Repower - Phase 1, Repower - Phase 2). An **Entity** is the legal owner, an LP such as SunE Alfred LP (378 Elgin and Manchester are entities in Repower - Phase 1). A **Site** is a place within an entity that has several, which today means only Firelight Solar's rooftops. A **Location** is something else: a physical place where stock sits (Oshawa, the Barn, the Office, or an entity's on-site shelf). The ticket system uses the same words.

1 Master List

Everyone can view and export **Inventory** edits

One row per part. Columns: GCS P/N, **Pic** (a reference picture), **Data** (datasheet), Manufacturer, Mfr Part #, Description, Category, **Used By** (the entities that require it) and Last Cost. Filter any column, and use `Clear filters` to start again.

Descriptions have three lines

1. **What it is**, a generic word (Relay, Contactor, Circuit Breaker, Fuse…).
2. **Model and ratings.**
3. **Anything else** (optional).

They are shown as three blocks: what it is in bold, then the model and rating, then the rest in grey. An older one-line description shows as plain text.

Inventory

- `Edit List` makes the rows editable. Edit a description in its three-line box. Press `Delete` on a row to remove a part, and save.
- **Picture:** in edit mode press **Attach** or **Replace**. The picture is shrunk automatically. Anyone can click the picture icon to see it.
- **Datasheet:** in edit mode each part has **PDF** (upload), **Link** (paste an address) and ✕ (remove). A new part must be saved before you can attach a picture or datasheet. Anyone can click the document icon to open it.
- `Import List`: a CSV with columns GCS P/N, Mfr Part #, Manufacturer, Category, Description. A row with an existing GCS P/N updates that part; a row with the P/N left blank adds a new part. **A column your file does not have is left alone**, so a file with only GCS P/N and Description changes only descriptions. You see a preview ("Add new" / "Update", unchanged cells marked) before anything is saved. Put multi-line descriptions in a quoted cell.
- `Export List` downloads the list as a CSV that Import can read back.
- `Change History` shows every change ever made to the Master List (who, when, old and new values) with filters.

The Part ID column is no longer shown. New parts still get one generated automatically.

2 Required Inventory

Inventory edits

Pick an **Entity**. The table lists the parts that entity requires: GCS P/N, Part ID, Manufacturer, Description, **Target Stock** (how many it should keep) and **Shared**.

- **Shared** means the target is a pool several entities draw on, not each entity's own. When totals are added up, shared targets count once (the largest) rather than being added together.
- `Edit List` (Inventory role) lets you tick parts on or off, set target stock and the Shared marks, with a search box to narrow the list. Press `Save Changes`.
- **Why it matters:** an entity can only order a part on a PO (as a spare or used part) if the part is on its Required Inventory list, and only entities with the part on their list can receive it in a transfer.

3 Inventory On Hand: Ownership

Everyone can view **Inventory** changes counts

A grid of parts down the side and entities across the top, showing how many each entity **owns**, with the target stock and the total owned. Choose an entity in the dropdown to see just that one, and `Export CSV` to download it. Click a part to jump to where it physically is.

Changing a count

Inventory

Press `Update Count`. You have three ways to change ownership.

1. **Individual Adjustment.** Press `Individual Adjustment`, type new numbers into the cells (search to narrow the list), write a **reason**, and press `Save Adjustment`. A window then asks **where the units are**, one row per changed count: for an increase, where the extra units are put; for a decrease, where they are taken from. Choose a location on every row (there is a *Set every row to* shortcut), then `Confirm & Save`. Places that do not hold enough units are greyed out, and *Automatic* takes from the entity's on-site shelf first, then Oshawa, the Barn, then the Office. An entity with no on-site shelf cannot be given an on-site location.
2. **Upload a count.** A CSV with columns GCS P/N, Entity and Quantity (a file that still says Site instead of Entity is accepted). Every row replaces that part's count for that entity, and you see a preview (previous and new) before confirming. One bad row cancels the whole upload. Extra units go on the entity's on-site shelf (Oshawa for an entity without one); removed units come off its on-site shelf first, then Oshawa, the Barn, then the Office. `Export Current Counts (CSV)` gives you a starting file.
3. **Stock Transfer.** Moves ownership from one entity to another and moves the units with it (see below).

Stock Transfer

Inventory

- Press `Stock Transfer`, choose **From** and **To** entities, type a reason, and say where the units are being taken from and where they end up (the sender's on-site shelf, Oshawa, Barn or Office; the receiver's on-site shelf only if it has one). In the table, **On site / off** shows how many of the sender's units are on its on-site shelf and how many are in Oshawa, the Barn or the Office. Type a quantity on the part's row and press `Transfer`.
- The move is treated as having already happened, so the owner and the location change together.
- The receiving entity must have the part on its Required Inventory list, or the Transfer button is off.
- **Each transfer also creates a PO** from the sender to the receiver, using the part's Last Cost as the price. It starts at the *Requested* stage and is taken through approval, issue, receiving and closing by hand like any other PO. Its number is shown on the History entry and in the confirmation message.

4 Inventory On Hand: Physical Location

Everyone can view **Inventory** moves and uses parts

A grid of parts against places: Oshawa, Barn, Office, then each entity that has an on-site shelf, with a total. There are three ways to narrow it, and they work together:

- The **search box** ("Search part number, description, maker..."). Every word you type must appear in the part number, Part ID, description, manufacturer or manufacturer part number. It also works while you are moving or using parts.
- The **location dropdown** shows only parts that have units in that place. `Export CSV` follows it, so with Oshawa chosen you export just what is in Oshawa.
- The **GCS P/N** dropdown in the heading.

Inventory

Update Location (ownership does not change)

- Type a reason (needed first). On the part's row choose **From** (where the units are now, with the count), **To**, a **Qty**, and press `Move`.
- An entity's on-site shelf can only receive units that entity owns. Options that cannot work are greyed with the reason, for example *(transfer ownership first)*, *(owns none off-site)* or *(no on-site shelf)*.
- Parts cannot go straight from one entity's on-site shelf to another's. Use Stock Transfer for that.

Record Part Use (the part was used)

- Type a reason, choose **Used from** (the place), and **Whose count** it comes from (automatic if it is an entity's on-site shelf; if it is Oshawa, the Barn or the Office, choose which owner), a quantity, and press `Record Use`. The units leave the place **and** that owner's count.

While you are moving or recording use, the yellow "on order" numbers are hidden, because these actions work on real stock only.

5 Inventory On Hand: History

Every stock change, one entry each: who, when, the reason, and for each part the before and after, shown as *Owned by …* or *Location: …*. Filter by type (adjustment, count, transfer, use, PO received, location move, opening balance) and press `Refresh`. History cannot be edited. The entry for a transfer names the PO that was created; the entry for receiving names the PO and vendor.

6 How stock goes up when a PO arrives

- Stock rises only when the **requisitioner** sets the PO's **Parts Status** to **Received**. Issuing the PO does not add stock, and neither does matching an invoice.
- If the PO has **spare** parts, a window asks where the units are (Oshawa, Barn, Office, or the entity's on-site shelf if it has one). They are added to that entity's count at that place, once.
- Used-immediately parts, consumables and services never add to stock.
- A pre-paid PO with spare parts must still be set to Received before it can be closed.

7 The other tabs, briefly

- **Suggestions.** Choose the type, the page it concerns, write your suggestion, and send it. You see its status and any reply on your own list. Admins see everyone's, with a New badge, a status (New, Looking at it, Planned, Done, Won't do), a reply box and an export.
- **Budget** (admin only for now). Pick the Project and Year. Each entity has its Budget, Issued POs, Actual Invoiced, To be invoiced and Remaining, by category (a **Remaining summary**

view shows % used). Click a figure to list the POs behind it. Only an admin can use `Edit budget`.

- **Audit** (admin only for now). A preview of an accounting comparison, on made-up data for now. Nothing it shows is saved.
- **My Signature** in the page header: draw or upload your signature once and it prints on the POs you approve.

Quick reference

I WANT TO...	GO TO	NEEDS
See how many of a part an entity owns	Inventory On Hand → Ownership	Sign in
Find where a part physically is	Inventory On Hand → Physical Location (use the search box)	Sign in
Fix a count	Ownership → Update Count → Individual Adjustment	Inventory
Load a recount from a spreadsheet	Ownership → Update Count → Upload a count	Inventory
Move a part to another place	Physical Location → Update Location	Inventory
Record that a part was used	Physical Location → Record Part Use	Inventory
Give a part to another entity	Ownership → Stock Transfer	Inventory
Add or change a part	Master List → Edit List or Import List	Inventory
Change what an entity needs	Required Inventory → Edit List	Inventory
Add parts to stock from an order	Purchase Orders → the PO → Parts Status → Received	The PO's Requisitioner
See who changed what	Inventory On Hand → History; Master List → Change History	Sign in

When something is refused

MESSAGE (IN SHORT)	WHAT TO DO
A reason is required	Every change needs a reason typed first.
A site can only hold parts it owns (the database message says "site" for an entity's on-site shelf)	Change ownership first (Stock Transfer or a count), then place the units.
… has no site of its own — choose Oshawa, the Barn or the Office	That entity (Firelight) keeps no on-site shelf. Pick another place.
Only N in that location	Pick the place that actually holds them, or move them there first.
… doesn't use this part — add it on Required Inventory first	The receiving entity must list the part before it can be transferred to it.
Not enough owned by …	The sender does not own that many.
Ownership and physical locations would no longer match	Something is off in the counts; fix the locations first (an inventory editor can use Update Location or an adjustment).

The other guides: Purchase Order Guide · Ticketing Guide

Written from how the system works in November 2026. If a button or message differs from this guide, the screen is right: tell an admin.